IB MATHEMATICS SL • UNIT 8.01

C Topic 8: IB SL Paper 1 & Paper 2 Extended-Response Synthesis

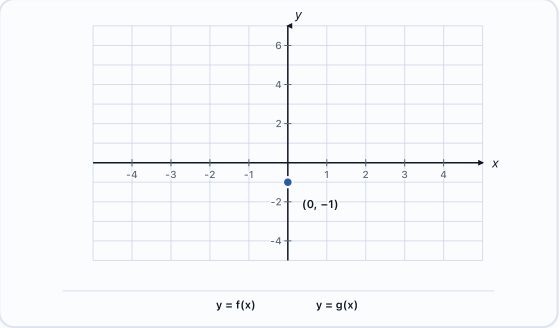
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 8: IB SL Paper 1 & Paper 2 Extended-Response Synthesis Practice Exercises

1 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?



A $(0, -3)$

B $(0, 3)$

C $(-1, 0)$

D $(3, 0)$

2 MCQ

Consider the function $f(x) = 2x + 3$. What is the value of $f(4)$?

A 7

B 11

C 10

D 5

3 MCQ

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

A 3

B 6

C 9

D 0

4 MCQ

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

A (3, 4)

B (-3, 4)

C (3, -4)

D (4, 3)

5 MCQ

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

A 5

B 6

C 7

D 4

6 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(0)$.

A 1

B -1

C 3

D -3

7 MCQ

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

A 6

B -3

C 3

D -6

8 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

A (1, 0)

B (3, 0)

C (0, 3)

D (0, 1)

9 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

A $y = 2x - 5$

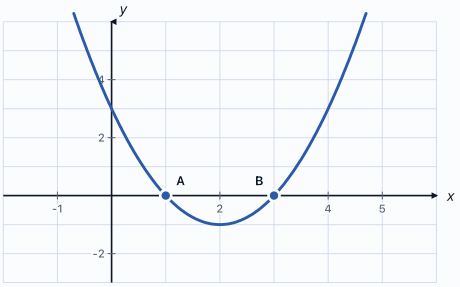
B $y = -\frac{1}{2}x + 5$

C $y = 2x - 8$

D $y = -2x + 11$

10 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

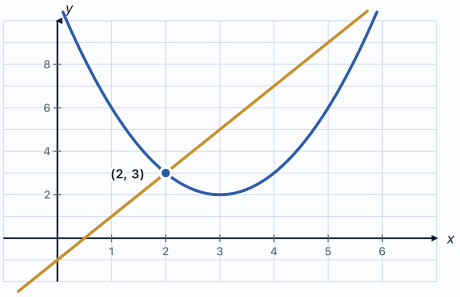
B 3

C 2

D 4

11 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

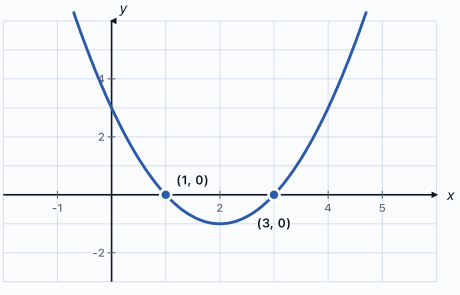
B 4

C 5

D 6

12 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.



A 3

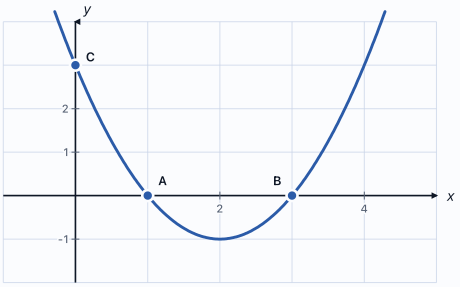
B 4

C -4

D 1

13 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.



A -4

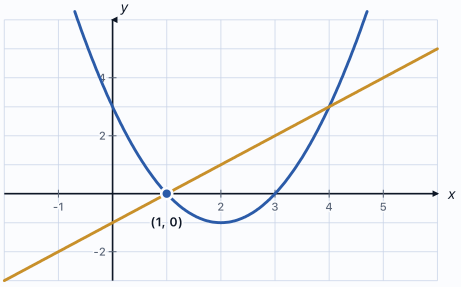
B 4

C 3

D 1

14 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.



A 1

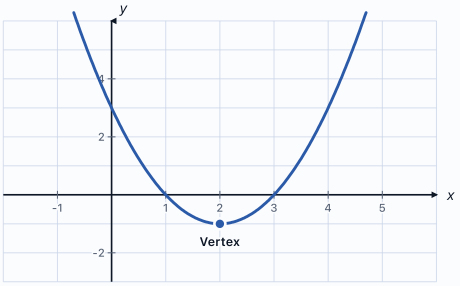
B 8

C 0

D 18

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.



A 2

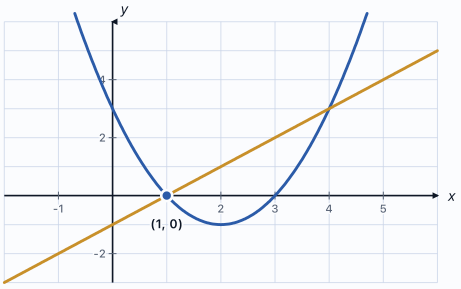
B -2

C 1

D 3

16 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



A (3, 2)

B (5, 4)

C (2, 1)

D (4, 3)

17 MCQ

$\$1200$ is invested at a simple interest rate of 8% per year for 5 years. Find the interest earned.

A 480.00

B 470.00

C 490.00

D 500.00

18 MCQ

Find the midpoint of $A(-1, -1)$ and $B(-7, 5)$.

A $(-6, 6)$

B $(-3, 2)$

C $(-4, 3)$

D $(-4, 2)$

19 MCQ

Find the midpoint of $A(1, 2)$ and $B(5, 10)$.

A $(3, 6)$

B $(3, 7)$

C $(4, 8)$

D $(4, 6)$

20 MCQ

$\$500$ is invested at a simple interest rate of 6% per year for 5 years. Find the interest earned.

A 160.00

B 150.00

C 170.00

D 140.00

21 MCQ

Calculate the midpoint of $A(8, 7)$ and $B(2, 13)$.

A $(-6, 6)$

B $(6, 10)$

C $(5, 11)$

D $(5, 10)$

22 MCQ

Find the midpoint of $A(1, -6)$ and $B(-9, 2)$.

A $(-4, -1)$

B $(-4, -2)$

C $(-10, 8)$

D $(-3, -2)$

23 MCQ

Find the midpoint of $A(-3, -8)$ and $B(-7, -10)$.

A $(-4, -9)$

B $(-4, -2)$

C $(-5, -8)$

D $(-5, -9)$

24 MCQ

$\$2000$ is invested at a simple interest rate of 2% per year for 3 years. Find the interest earned.

A 110.00

B 140.00

C 120.00

D 130.00

25 MCQ

$\$500$ is invested at a simple interest rate of 4% per year for 2 years. Find the interest earned.

A 30.00

B 40.00

C 50.00

D 60.00